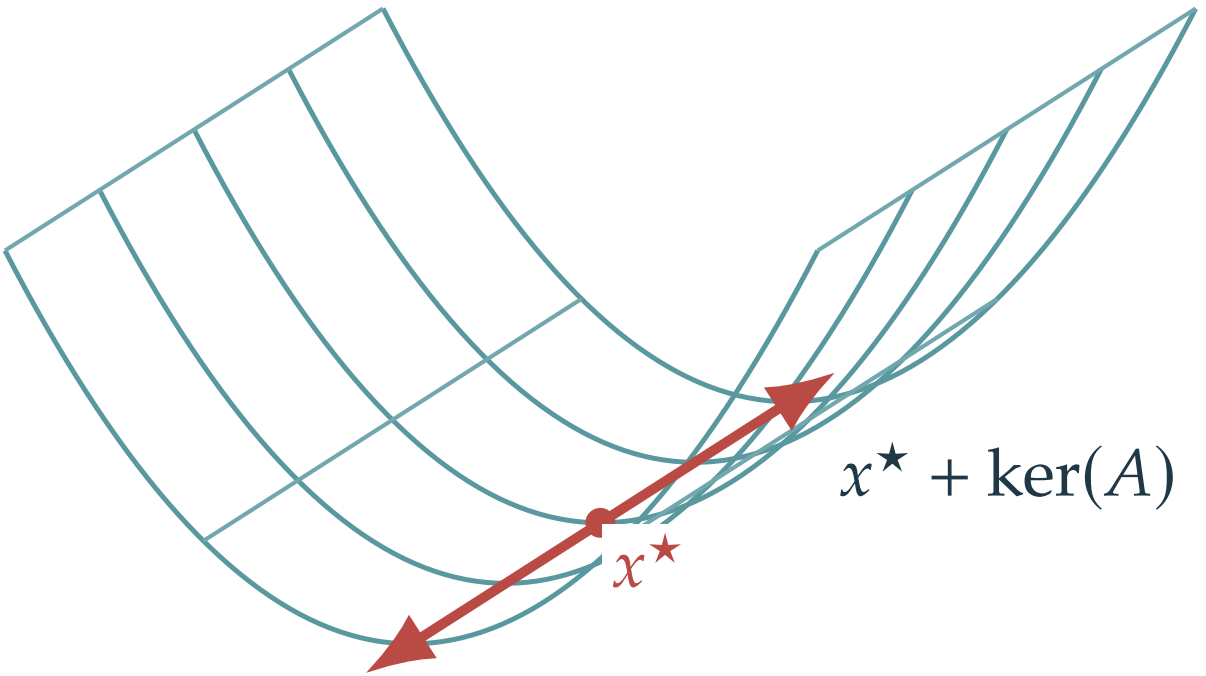
$\ker(A) \neq \{0\}$



Unbounded set of minimizers